

Moving ahead with small modular reactors

The joint statement issued during the official state visit of Prime Minister Narendra Modi to the US in late June 2023 had this to say about nuclear co-operation and small modular reactors (SMRs): “Prime Minister Modi and President Biden underscored the important role nuclear energy plays in global decarbonisation efforts and affirmed nuclear energy as a necessary resource to meet our nations’ climate, energy transition, and energy security needs.... They also noted the ongoing discussion on developing next generation Small Modular Reactor technologies in a collaborative mode for the domestic market as well as for export. The United States reaffirms its support for India’s membership in the Nuclear Suppliers Group and commits to continue engagement with likeminded partners to advance this goal.”

Politically, it is a significant statement, coming as it does 18 years since the announcement of the historic US-India Civilian Nuclear Agreement. This high-profile deal that was signed in 2005 by then Prime Minister Manmohan Singh and US President George Bush involved bitter political wrangling at that time.

Now a strategy is emerging for India to move from larger to smaller nuclear plants. The NITI Aayog has published a report titled “The Role of Small Modular Reactors (SMRs) in Energy Transition.” In his address at a workshop on SMRs, Jitendra Singh, Union Minister of State for Atomic Energy (as part of his larger portfolio), said India would be taking steps for development of SMRs with up to 300 MW capacity to fulfil its commitment to clean energy transition. He emphasised the government’s keenness to involve private sector participation in this initiative.

What are SMRs? They are advanced nuclear reactors that have a capacity of up to 300 MW per unit, which is about one-third of the generating capacity

of traditional nuclear power reactors. According to the International Atomic Energy Agency, they make it possible for systems and components to be factory-assembled and transported as a unit to a location for installation. Besides offering significant savings in cost and construction time, SMRs are simpler and safer compared to larger nuclear plants. SMRs also operate longer without the need for refuelling. While traditional nuclear plants require refuelling in 1 to 2 years, SMRs need refuelling every 3 to 7 years.

There are more than 70 SMR designs under development in 17 countries. Several countries have already invested heavily in SMR technology. Recently the United States Nuclear Regulatory Commission approved the design of NuScale Power company for a plant of 600 MW capacity consisting of 12 modules of 50 MW SMRs. A few of them are expected to be commissioned before 2030.

In Russia, an SMR, named Akademik Lomonosov, having a capacity to produce 77 MW power, was commissioned as early as 2019. Rosatom of Russia is also building a reactor based on heavy metal coolant

with a 100 MW capacity. Recently, China commissioned a high-temperature gas-cooled modular pebble bed demonstration SMR.

In the United Kingdom, Rolls-Royce is establishing a centralised manufacturing facility to produce SMRs. In Poland, one electricity utility, in collaboration with a US company, has decided to install SMRs on its site after retiring the old coal-powered plants, making effective use of the available resources like grid connections, water supply, civil structures and skilled workers.

NITI Aayog’s strident suggestion for involvement of the private sector is a new twist to an area that has been reserved for the public sector. The report advocates incentives for the private sector for setting up SMRs across the country in public-private partnership (PPP)

mode. Stakeholder consultations to firm up a policy framework are already underway. The NITI Aayog paper also sets out the contours of a regulatory regime, including strident safety standards and regular monitoring. It also recommends changes to India’s foreign investment policies so that both domestic and foreign private companies can participate. Several foreign companies, including Westinghouse Electric, GE-Hitachi, Electricite de France and Rosatom, have evinced interest in participating in the country’s SMR opportunity.

One important point that emerged during the 2021 United Nations Climate Change Conference (COP-26) at Glasgow was that renewable energy resources such as solar, wind and hydropower cannot always be relied upon to supplement fossil fuel-generated energy as they are weather-dependent. In a recent article, published by the Centre for Social and Economic Progress, the case for increasing nuclear share in India’s decarbonising energy strategy is well laid out. It mentions: “On balance, there is a good case for expanding the share of nuclear power in India. China is targeting a 10 per cent share of nuclear power in its energy mix by 2035.”

The Central Electricity Authority, in its recently released National Electricity Plan (NEP) 2022-32, notes that as of March 31, 2022, Nuclear Power Corporation India Limited operated 21 reactors with an installed capacity of 6,780 MW, which is 1.7 per cent of the total installed power capacity in the country. The NEP goes on to forecast that in 2032, India would have 19,700 MW of Nuclear capacity representing 2.2 per cent of capacity and 4.4 per cent of gross generation.

If India aspires to raise nuclear power to 10 per cent of its energy-mix around 2035, then the nuclear capacity would have to be close to 90GW.

Perhaps, a proliferation of SMRs could aim to pursue that target!

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